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Prof. Nicolás Quesada

Polyquantique Research Group

École Polytechnique de Montréal, Canada

Re: Motivation Letter — Non-Gaussian PhD Applicant

Dear Prof. Quesada,

I am writing to express my strong interest in the PhD position in theoretical quantum optics at your Polyquantique Research Group. I have recently completed my MS in Physics at Quaid-i-Azam University, Islamabad, Pakistan, with a thesis titled “*Generation of Gaussian Quantum Steering and Entanglement in Coupled Lossy Waveguides*.” My research lies at the intersection of Gaussian quantum optics and integrated photonics, aligning directly with your group’s focus on optical parametric amplifiers (OPAs) as building blocks for quantum communication networks.

In my thesis I modelled a two-mode coupled lossy optical waveguide system using the covariance matrix formalism and Lindblad master equations. I derived closed-form analytical solutions for the second-order operator moments and used them to quantify entanglement via logarithmic negativity (E_N) and steering via the Gaussian steering measure ($\mathcal{S}$). Through a systematic parameter study I investigated how the loss rate κ , nonclassicality τ , purity μ , coupling strength J , and squeezing r govern quantum correlations during propagation. A central result was that J partially compensates the adverse effects of κ , preserving useful correlations over longer propagation lengths. I further confirmed the strict quantum hierarchy (Gaussian Steering $\subsetneq$ Gaussian Entanglement) across all parameter regimes, including squeezed-state inputs under realistic loss.

I was particularly drawn to your work on simulating lossy and partially distinguishable quantum optical circuits (arXiv:2412.17742), where noise treatment through mixed Gaussian states connects directly to my study of how κ degrades photonic quantum correlations. My background in Gaussian-state dynamics and covariance matrix methods is also relevant to your NSERC project on classical simulation of near-term photonic quantum computers. While my thesis operates in the Gaussian regime, I am eager to extend this foundation into the non-Gaussian domain—developing analytical and numerical tools to study OPAs beyond squeezing, which is precisely the focus of the advertised PhD.

Beyond my theoretical background, I bring solid scientific computing skills developed through implementing covariance matrix evolution, symplectic eigenvalue computations, and parametric simulations in Python. I am comfortable working at the interface of analytical theory and numerical methods, and I am genuinely committed to your group’s culture of open-source software and reproducible science.

I am aiming to begin my PhD in 2026 and am excited by the prospect of contributing to Polyquantique’s collaborative projects with government and academic laboratories. I would welcome the opportunity to discuss my background with you at your convenience.

Yours sincerely,

Hafsa Zia

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